CV

AI CAD-to-Cost Analysis

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AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

20

Seat Locking Bracket

256 × 226 × 31 mm · 71.1 cm³ · Al 0.192 kg / Steel 0.558 kg

Manufacturability: 20/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	**255.8 × 226.1 × 31.0 mm**	Surface Area	**919.4 cm²**
Volume	**71.07 cm³**	Weight (Al)	**0.192 kg**
Weight (Steel)	**0.558 kg**	Weight (Plastic)	**0.075 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Bend lines	15	High	Multiple formed bends consistent with 15-bend progressive die sequence
Free-form / B-spline surfaces	235	High	235 of 426 faces are B-spline — unusually organic for sheet metal, suggests wrap-formed flanges or compound curvature requiring multi-slide/progressive forming stations
Threaded features	1	Medium	Threaded holes/inserts detected — will require secondary tapping, self-clinch fastener insertion, or PEM nut station
Undercuts / re-entrant features	32	High	32 undercut faces relative to Z draw direction — indicates side-formed flanges or hemmed edges needing extra die actions
Thin-gauge zones	1	Medium	Min wall 0.53mm vs mean 1.55mm — local thinning risk at tight bend radii

§3 — Material Analysis

AI-suggested

Primary Material: **High-Strength Steel (HSLA, ~1.5-2.0mm) (62% confidence)**

Part is a structural 'seat locking bracket' — a safety-critical automotive interior/seat-structure component that must resist occupant-load and crash forces without failing at the latch. Such brackets are standard high-strength/HSLA steel stampings, not aluminium or mild steel, despite the thin (0.53-2.17mm) measured wall — thin gauge here reflects sheet thickness, not a casting wall. User-forced process is sheet_metal, and material ground truth was not supplied, so steel (denser, load-bearing, EN-standard HSS) is selected per guidance to default to steel for structural automotive brackets unless an aluminium signal is present.

Alternatives: **DC01 Cold-Rolled Mild Steel (25%) · Aluminium 5052 (if weight-critical variant) (10%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Progressive die stamping (blank + 15 bend/form stations)	sheet_metal	90%	0.0005	User-forced sheet_metal process; part geometry (thin non-uniform wall 0.53-2.17mm, bend-dominated topology, large flat blank envelope 269x237mm) is consistent with a multi-station progressive stamping tool at 200,000 units/yr volume.

§5 — Manufacturability Risks (Score: 20/100)

Severity	Feature / Area	Description	Recommended Action
High	**32 undercut / re-entrant faces**	Undercuts relative to the vertical draw direction imply formed flanges or hems that cannot be produced in a single straight-draw progressive station.	Add cam-actuated side-forming or flanging stations, or split into two sub-assemblies joined by spot-weld/rivet.
High	**235 B-spline (free-form) faces**	Very high proportion of free-form surface for a stamped bracket suggests compound/wrap curvature not typical of simple sheet forming; risk of springback and wrinkling at tight radii.	Validate formability with FE simulation (AutoForm/PamStamp); add draw-in beads and trim allowance to control springback.
Medium	**Min wall 0.53mm**	Local thinning at bend/draw radii below mean gauge risks tearing or fatigue cracking at the locking interface.	Increase local bend radius or add local material reinforcement (doubler/gusset) at the thin zone.
Medium	**Threaded feature**	Detected thread requires a secondary tapping or self-clinching fastener insertion operation not covered by the stamping cycle.	Specify PEM/rivet-nut insertion press station in-line, or post-process tapping cell.
Low	**15 bend stations**	High bend count increases die complexity and stack tolerance risk across a 12-station progressive tool.	Group non-critical bends into common stations and verify cumulative tolerance stack against locking-feature datum.

§6 — DFM Issues (sheet_metal)

Severity	Area	Description	Impact	Fix
High	**Formed undercuts (32 faces)**	Multiple re-entrant/undercut geometries relative to the press draw axis cannot be formed in a straight vertical stroke.	Requires extra cam/side-action stations or secondary forming press, adding 15-25% to die cost and cycle time.	Re-orient part or split flange features so all forming is achievable with straight punch motion; use cam-driven slides for unavoidable undercuts.
High	**Thin-gauge bend zones (min 0.53mm)**	Wall thickness thins locally at tight-radius bends, well below the 1.55mm mean, risking cracking at the locking tab.	Scrap/rework rate increase and potential in-field fatigue failure of the locking feature.	I n c r e a s e b e n d r a d i u s t o "e 2 x material thickness at critical bends and add local reinforcement rib.
Medium	**High bend count (15 bends / 12 stations)**	Complex progressive die with many forming stations increases cumulative dimensional stack-up on the locking geometry.	Tolerance drift can cause latch mis-fit, driving inspection/rework cost.	Consolidate bends where possible and add a calibration/restrike station near the critical locking feature.
Medium	**Threaded feature**	In-die thread forming is impractical; a threaded feature detected on a stamped part implies a secondary tapping or fastener-insertion step.	Added secondary operation cost and cycle time per part (~0.01hr).	Use a self-piercing/self-clinching rivet nut inserted in-die rather than post-process tapping.

Severity	Area	Description	Impact	Fix
Low	**Large blank envelope (269x237mm)**	Large flat blank relative to part height gives a low material utilisation ratio, increasing scrap per part.	Higher steel cost per part at high (200k/yr) volume.	Nest blanks in strip layout to minimise web/scrap and consider double-row nesting where geometry allows.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£1.05	£1.65	£2.60

Net Weight	**0.558 kg**	Material	**mat-hss**
Recommended Process	**sheet_metal**	Cycle Time	**0.0005 hr/part**
Setup Time	**1.500 hr**	Operations	**2 ops**
Operations Detail	**Progressive stamping (blank+form+bend, 12 stations) (mach-haas-vf2, 0.0005 hr) Tap/insert threaded feature (mach-drill, 0.0100 hr)**		

Process-Specific Parameters

Casting Subtype	**sand**
Die/Mould Cost	**£0**
Die Life	**0 shots**
Cavities	**0**
Yield	**0.0%**
Flash Weight	**0.000 kg**
Yield	**0.0%**
Die Cost	**£0**
Strokes	**0**
Cavities	**0**
Wall Thickness	**0 mm**
Mould Cost	**£0**
Mould Life	**0 shots**
Projected Area	**0.0 cm²**

§8 — AI Analysis Explanation

The Seat Locking Bracket is analysed as a progressive-die sheet-metal stamping per the forced classification. Geometry (thin 0.53-2.17mm walls, 1054 mostly line/bspline edges, and a large flat 255x226mm envelope) is consistent with a stamped/formed bracket rather than a casting, despite the low fill ratio which is an artefact of the thin-shell sheet geometry. Because it is a seat-locking structural bracket (safety-critical latch component), high-strength steel (mat-hss) was selected over aluminium per the material default rule, giving a net part weight of 0.558kg. The 12-station (1 blank + 15 bend) progressive die costing £25,000 (geometry-derived) amortises efficiently across the 200,000 unit/year volume. Key manufacturability concerns are the 32 undercut faces (needing cam-actuated forming) and the unusually high free-form surface content (235/426 faces), both of which increase die complexity beyond a typical flat bracket stamping.

§9 — Analysis Limitations & Assumptions

1. Material (steel vs aluminium) and sheet thickness were not confirmed by drawing/photo — inferred from part function (safety-critical seat lock) per steel-default guidance.
2. High free-form (B-spline) surface content is unusual for sheet metal and may indicate the STEP model includes tessellated/organic surfaces not representative of a true flat-pattern stamped part; recommend reviewing the CAD source.
3. Progressive die life and setup time are rule-of-thumb estimates for HSS at this gauge, not geometry-derived.
4. Threaded feature detected but exact size/location not resolved from hole-radius data (all reported hole radii were below detection threshold).

Stage-1 pre-selection: sheet_metal (100%) ·